MASTER'S THESIS PROPOSAL PRESENTATION

A Methodology for Transforming Natural Language Academic Policies into Formal Knowledge for Automated Reasoning

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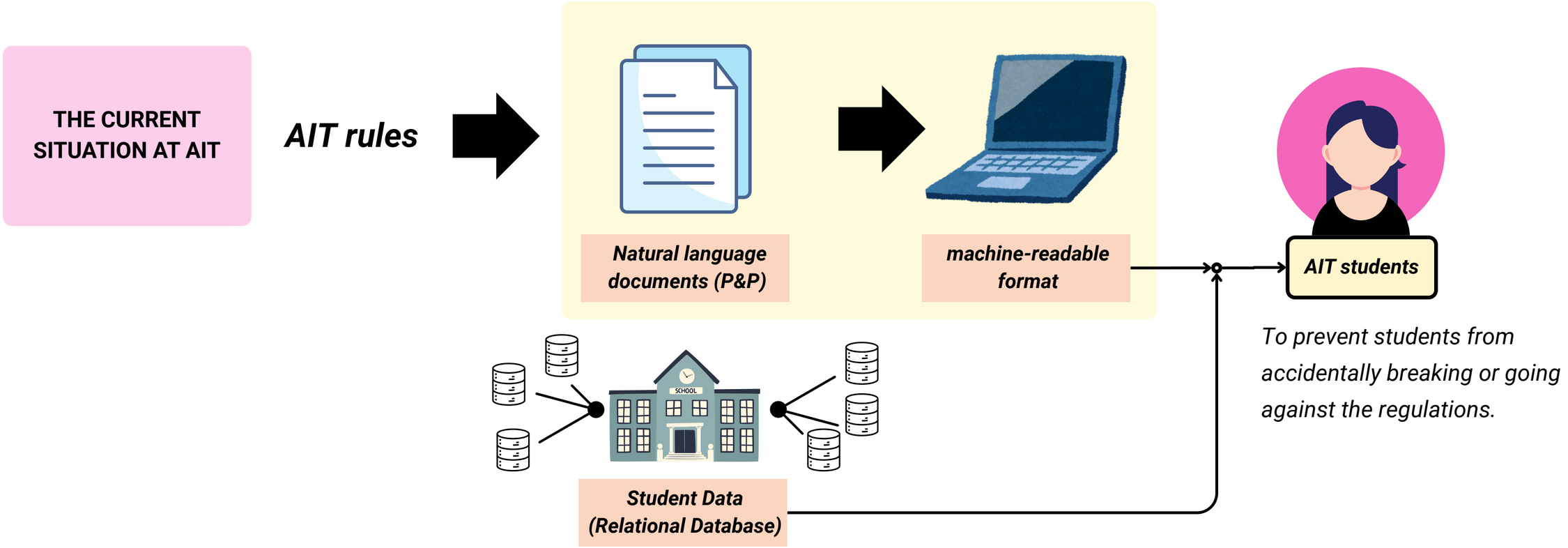
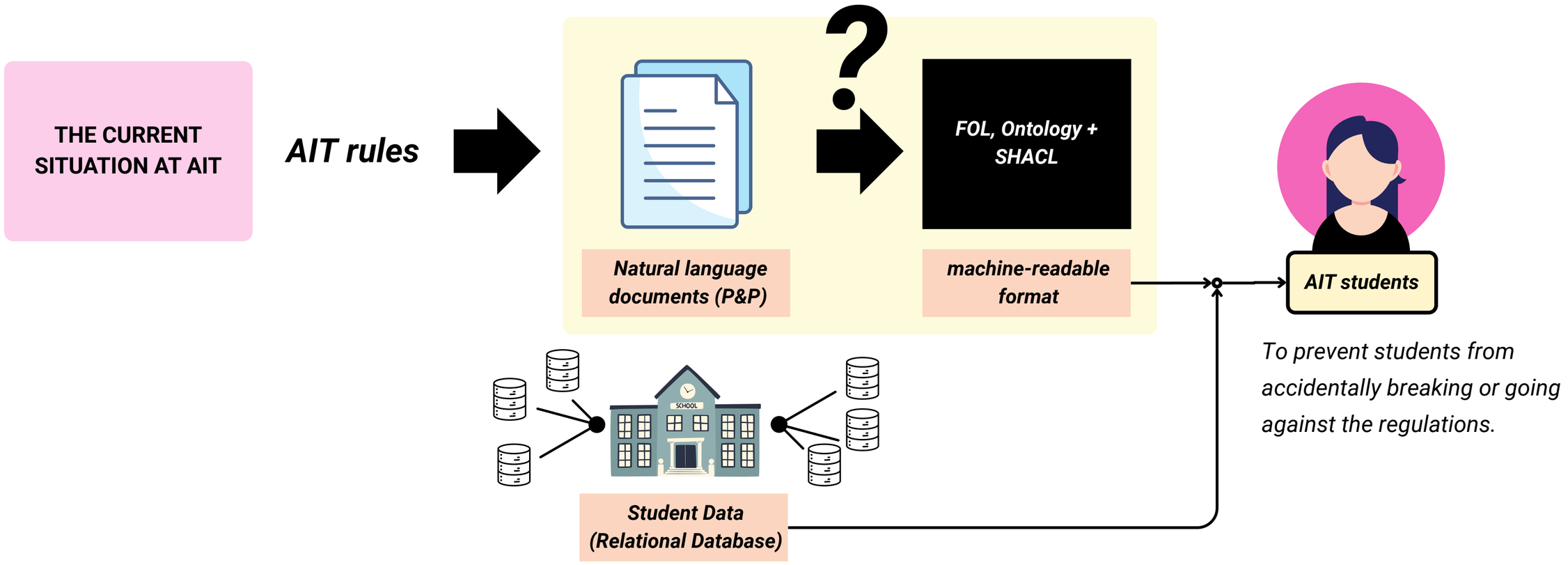
INTRODUCTION

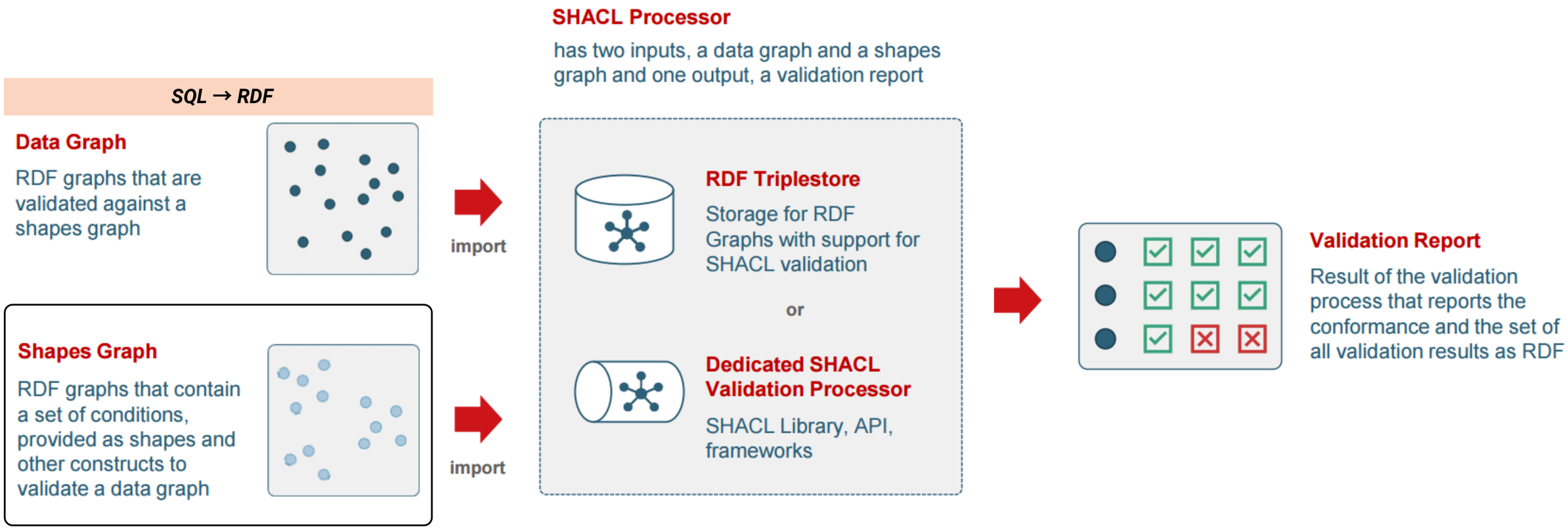


INTRODUCTION



To prevent students from accidentally breaking or going against the regulations.

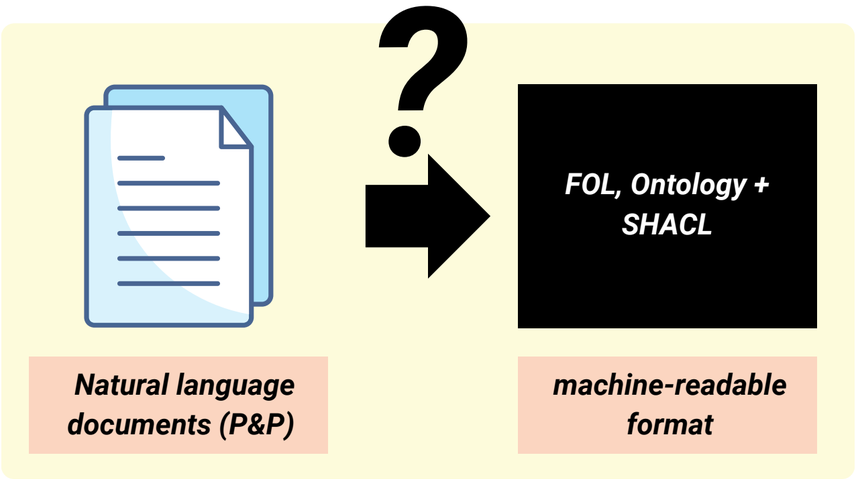
HOW DOES IT WORK - SEMANTIC WEB TOOLS



INTRODUCTION PROBLEM STATEMENT

The Core problem:

The lack of a systematic way to transform the AIT policy into semantic web environment.



- *We propose a framework/guideline for translating a policy into **FOL statement** that is compatible with **Ontology** and **SHACL***

INTRODUCTION

Summarized Methodology

1. ***Classify AIT policies*** *that are rated as complexity and clarity levels*
2. ***Define FOL syntax*** *of each policy class.*
3. ***Translate*** *each policy statement to* ***FOL statement***
    a. *Simplify policy statement*
    b. *Define the vocabulary (variables, predicates)*
    c. *Translate the policy into FOL statement*
    d. *Analyze using Assumption-Based Argumentation (Dung, 1995, 2009)*
4. ***Convert*** *the component from FOL statement into roles in* ***Ontology*** *and* ***SHACL*** *which it's called* ***Shape Graph***
5. ***Evaluation/Inferencing***
    a. *Fetch a student data from database and convert it into* ***Data Graph***
    b. ***Validate the context*** *which is a student data based on crafted ontology using SHACL*
    c. ***Validation report will be generated*** *once the contradiction is found.*

Contribution

1. *Propose a framework/guideline to translate policy statement into FOL statement*
2. *Propose a framework/guideline to convert FOL statement to Ontology and SHACL*
3. *Demonstrate that FOL statement can be implement to model a policy compliant checking using Ontology and SHACL.*

RESEARCH QUESTION

- ***RQ1:*** *What constitutes a systematic methodology for transforming academic regulations from natural language policies into a formal knowledge base,* ***utilizing First-Order Logic (FOL) as an intermediate representation and Semantic Web technologies (Ontology, SHACL) for implementation****?*

HYPOTHESIS

- ***H1 :*** *A systematic methodology using* ***FOL will be able to formalize even complex, multi-conditional rules*** *into a* ***consistent, machine-verifiable format.***

- ***H2:*** *The knowledge base created by this methodology will be sufficiently robust to* ***serve as the foundation for a functional automated reasoning application****, such as a proactive compliance checker.*

Scope

- *Policy Classification is done by hand.*
- *Due to time constrain, we start with the rules are rated as **"Clear"** according to the definition.*

OBJECTIVES

- *To design and propose a **systematic methodology** for the analysis, formalization, and modeling of academic regulations from unstructured P&P documents.*
- *To define a process that **uses First-Order Logic (FOL) as an intermediate layer** to capture the precise logic of a rule before its implementation in a specific technology, such as Ontology, SHACL.*
- *To demonstrate the feasibility and effectiveness of the proposed methodology by developing a **Proof-of-Concept (POC) knowledge base using selected AIT policies.***

LITERATURE REVIEW

Research paper in section of FOL analysis

Research Paper	Pros	Cons	Techniques	Evaluation	Gap Analysis & Our Positioning
Assumption-Based Argumentation • *(Dung, 1995, 2009)*	**It provides a strong theoretical foundation.** It shows that human reasoning and conflict resolution can be modeled using abstract mathematics.	**It is highly theoretical and abstract.** It does not offer a concrete methodology or the technical tools to apply these concepts to real-world problems directly.	Abstract Argumentation Frameworks, Assumption-Based Argumentation (ABA), Formal Logic.	Theoretical and logical proofs. There is no evaluation with real-world data.	Our work makes this abstract theory practical. We apply the **"Acceptability of Arguments" principle to our Knowledge Modeling** process, *using Semantic Web technologies (Ontology, SHACL)* as the tools to implement this theory.

LITERATURE REVIEW

Research paper in section of Semantic Web Tools

Research Paper	Pros	Cons	Techniques	Evaluation	Gap Analysis & Our Positioning
A SHACL-Based Approach for Enhancing Automated Compliance Checking with RDF Data • *(Anim et al., 2024)*	It demonstrates **the power of SHACL-SPARQL to solve a very similar, complex problem** (aggregation & temporal rules) in a real-world scenario successfully.	The paper focuses mainly on proving the capability of the "language" itself. **It does not propose a complete "framework" or "system architecture"** that covers the entire process from rule analysis to user notification.	SHACL-SPARQL, Ontology Design Patterns (TVC), MeLOn Methodology.	An in-depth Case Study in the legal domain of the energy sector, proving that the approach is feasible and effective.	This paper's work is like a powerful engine, but **our work presents the "entire car."** *We provide a complete framework and system architecture*, which is a broader contribution.

LITERATURE REVIEW

Research paper in section of FOL + Semantic Web Tools

Research Paper	Pros	Cons	Techniques	Evaluation	Gap Analysis & Our Positioning
SHACL2FOL: An FOL Toolkit for SHACL Decision Problems • *(Pareti, 2024)*	It is the **first practical and automatic tool** that translates SHACL into First-Order Logic (FOL). It provides a **formally-grounded solution** to key decision problems like satisfiability and containment, bridging theory and practice.	The implementation is **not yet complete** for all complex SHACL constraints, especially those on literals. Scalability testing is **preliminary** and has only been performed on small-scale graphs.	First-Order Logic (FOL) SHACL Automated Theorem Provers (e.g., Vampire, E) TPTP format	**Preliminary functional testing** on small shape and data graphs to prove the concept. There is no comprehensive performance benchmark against other validators.	This paper provides a tool for **analysis,** translating from **SHACL → FOL.** This validates our choice of FOL as a sound intermediate representation. Our work addresses the opposite and more foundational problem: creating a **methodology for creation**, translating from **Natural Language → FOL → SHACL**. We are building the bridge that leads to the formal rules that a tool like this would analyze.

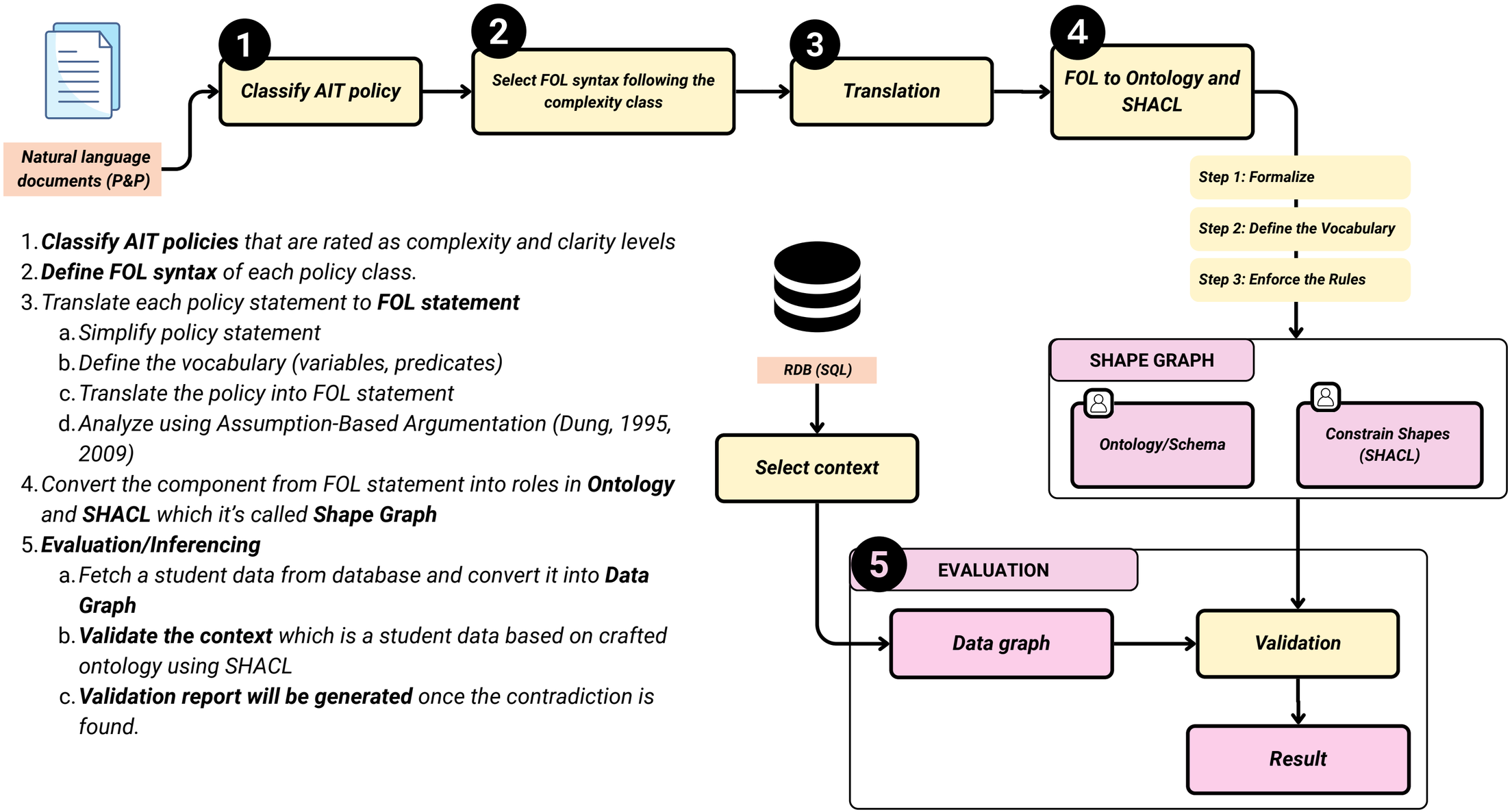
Dataset

- **Dataset:** This research uses ***five Policies and Procedures (P&P) documents*** from the Asian Institute of Technology (AIT) and an anonymized student database as the primary case study.
    - *AA-4-1-1 Academic Integrity in Research and Publication*
    - *FB-6-1-1 Credit Policy*
    - *FS-1-1-1 Campus Accommodation for Students*
    - *PA-2-1-2 Ethical Behavior and Grievance Process*
    - *Student-Handbook*

- **Regulations:** The research scope focus on only rules that related to the status of ***current students and rated as "Clear"*** which have 38 rules according to the definition

METHODOLOGY

PROPOSED SYSTEMATIC FRAMEWORK



1. ***Classify AIT policies*** *that are rated as complexity and clarity levels*
2. ***Define FOL syntax*** *of each policy class.*
3. *Translate each policy statement to* ***FOL statement***
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AIT P&P Document

AIT P&P Document

- *AA-4-1-1 Academic Integrity in Research and Publication*
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- *Student-Handbook*

1. **Complexity Level**
   - **Low:** Rules that are clear commands or prohibitions, ***simple and direct, with no extra conditions.***
     - ***Example:*** *"No pets allowed."*
   - **Medium:** Rules with ***1–2 conditions, exceptions, or short steps*** to follow.
     - ***Example:*** *"Cooking is not allowed, except in designated dorms."*
   - **High:** Rules that ***involve many steps, affect different areas*** (such as finance, housing, or visa), refer to other departments, or include complex penalties with different levels.
2. **Clarity Assessment**
   - **Clear:** The statement is clear, needs no interpretation, and can be followed immediately.
   - **Partially Ambiguous:** Mostly clear, but may include some terms or conditions that are vague and could lead to different interpretations.
     - *Example: the term "large electrical appliances" is not defined.*
   - **Ambiguous:** The statement is unclear, and the real intention cannot be understood without more information

METHEDOLOGY SELECTED RULES

Clear 38 rules in total from fives AIT P&P Documents

Rule ID	Cathegory	Details	Group	Complexity	Clarity	File name	Section / Heading	Page
R-001	Campus Accommodation	Students holding **a current AIT student ID card and registered** in the doctoral, master's, certificate, or special program are provided with accommodation on campus by the Accommodation Unit.	Student Rights & Services	Low	Clear	FS-1-1.pdf	IV. ELIGIBILITY	1
R-002	Campus Accommodation	Married Students are permitted to bring their immediate families (spouse and children) to join them during their studies. Married students and their immediate families can reside on-campus **when they have been allocated a married accommodation unit.**	Accommodation	Medium	Clear	FS-1-1.pdf	IV. ELIGIBILITY	2
R-003	Campus Accommodation	Graduating/Completing Students are required to vacate their rooms **within five days (including weekends)** after their graduation/completion of studies.	Accommodation	Low	Clear	FS-1-1.pdf	IV. ELIGIBILITY	3
R-004	Campus Accommodation	Cooking is not permitted, **except in accommodation units where cooking facilities or shared cooking facilities are provided.**	Accommodation	Medium	Clear	FS-1-1.pdf	VI. REGULATIONS	5
R-005	Campus Accommodation	The fine for **removal of furniture and appliances** is Baht 500 per item.	Discipline & Ethics	Low	Clear	FS-1-1.pdf	VII. SANCTIONS	6
R-006	Campus Accommodation	**Students who cook in dormitories where cooking is prohibited** may face a minimum fine of Baht 2,000, or the actual costs of the damage caused by cooking.	Discipline & Ethics	Medium	Clear	FS-1-1.pdf	VII. SANCTIONS	7
R-007	Campus Accommodation	**Pets are not allowed** in student accommodation. The fine for non-compliance is Baht 2,000.	Accommodation	Low	Clear	FS-1-1.pdf	VII. SANCTIONS	7
R-008	Campus Accommodation	Self-funded students are required to pay a one-month advance deposit. **For new students, the first invoice includes rental for two months.**	Finance & Scholarship	Medium	Clear	FS-1-1.pdf	VIII. RENTAL CHARGING SYSTEM	7
R-009	Campus Accommodation	Payments made **after 30 days** from the date of an invoice, **will generate an additional interest penalty** of 1.5% per month. **If the outstanding amount exceeds two months rent**, the student **will be asked to vacate the room.**	Finance & Scholarship	High	Clear	FS-1-1.pdf	VIII. RENTAL CHARGING SYSTEM	8
R-010	Ethical Behavior	This Ethical Behavior and Grievance Process **forms an integral part of the Terms and Conditions of study** of all students at AIT.	Discipline & Ethics	Low	Clear	PA-2-1-2.pdf	I. PREAMBLE, A. Applicability	1
R-011	Ethical Behavior	AIT employees, students and residents **must act in the interests of the Institute and avoid engaging in any actions contrary to the policies and interests of the Institute.**	Discipline & Ethics	Low	Clear	PA-2-1-2.pdf	II. ETHICAL BEHAVIOR, A. General policies	2
R-012	Ethical Behavior	**When expressing personal opinions outside the Institute**, AIT employees and students are required to **note explicitly that the opinions expressed are personal** and **do not necessarily represent the position of the Institute.**	Discipline & Ethics	Medium	Clear	PA-2-1-2.pdf	II. ETHICAL BEHAVIOR, B. Specific requirements	2
R-037	Credit Policy, Fees, and Invoicing	Dismissed' students lose their status as students. **Once dismissed**, students who are staying in campus accommodation **shall be asked to vacate their dormitory rooms; and their educational visas will be cancelled.**	Finance & Scholarship	High	Clear	FB-6-1-1.pdf	III. TUITION AND ALL OTHER FEES	3
R-038	Credit Policy, Fees, and Invoicing	Students with donor funding who **resign after the deadline** for adding and dropping of courses **will be liable to repay the total amount** of tuition and registration fees that have been charged to the donor.	Finance & Scholarship	Medium	Clear	FB-6-1-1.pdf	III. TUITION AND ALL OTHER FEES	4

POC - SELECT P&P RULES

- **Rule 1**

III. **TUITION AND ALL OTHER FEES**

1. For self-support students and holders of externally-managed scholarships (externally-supported/sponsored students who have their own funds):

(1) With regard to self-paying students and students with external sponsors, first semester/term fees must be paid in advance and/or fully paid upon enrolment, otherwise they will not be allowed to register. Consequently, the student ID and Internet/e-mail access will not be released by Registry if fees are not paid.

(2) Continuing full-time postgraduate (PG) students (Master, Doctor and Diploma), i.e., those who completed at least one semester/term, will be invoiced for the applicable registration, tuition, and other fees each semester/term from the second semester/term onwards, which must be paid in advance and/or fully paid before each semester/term starts, otherwise they will not be allowed to register.

(3) For students under the Professional Master program, they will be invoiced for the applicable registration, tuition, and other fees for the second semester/term. The invoice must be paid in advance and/or fully paid up before each semester/term starts, otherwise they will not be allowed to register.

- **Rule 2**

IV. ACCOMMODATION AND UTILITIES

A. Student Accommodation Rental

(1) Students are required to pay a security deposit, which is collected in advance.

(2) Housing rental is payable on the first day of each month.

(3) With regard to unpaid rental and utilities, interest of 1.5% per month is imposed if the amount remains unpaid after 30 days.

(4) If the outstanding amount exceeds two-month rental, the students will be required to vacate the rooms.

(5) Eviction letters will be sent by OFAM/Accommodation Unit to the concerned students with a grace period to pay the outstanding amount, which include interest. Otherwise, concerned students have to vacate the room within two weeks after the given deadline to pay.

(6) If accommodation and other fees remain unpaid beyond two months, a student's status shall be 'suspended/dismissed due to financial inability'. Viewing of grades and transcripts will be disallowed.

(7) Allocation of new rooms to students (based on queuing) will not be allowed for those with outstanding accommodation rentals.

- *FB-6-1-1: CREDIT POLICY*
  - ***Rule 1:*** *III. TUITION AND ALL OTHER FEES, Subsection 1. (2), page 1*
  - ***Rule 2:*** *IV. ACCOMMODATION AND UTILITIES, Subsection A. (6), page 5*

POC - TRANSLATING P&P RULES INTO FIRST-ORDER LOGIC (FOL)

3 *FOL statement translation*

3b

Variables

- **Student(x)**: x is a student.
- **Obligation(y):** y is a Obligation

Predicates
(Functional Predicate)

- **StudentType(x, t)**: x has the student type t (e.g., "Self-Support").
- **EnrollmentStatus(x, s)**: x has the enrollment status s (e.g., "Enrolled").
- **ObligationType(y, t)**: The obligation y is of type t (e.g., "TuitionFee", "AccommodationFee").
- **OutstandingDuration(y, d)**: The obligation y is overdue by d days.
- **Violation(x, r)**: x has a violation with the reason r.

Predicates
(Relational Predicate)

- **HasObligation(x, y)**: x has a financial obligation y.
- **IsPaid(y, b)**: The obligation y has a payment status of b (true/false).

Proposition
(logical statement)

- **Rule 1: Tuition Fee Compliance (TuitionComplianceShape)**
  - **3a** **P&P Statement:** "Continuing full-time postgraduate (PG) students... will be invoiced... which must be paid in advance and/or fully paid before each semester/term starts..."
  - **3c** **FOL Statement:** *"For all students x, **if** x is a 'Self-Support' student **and** has an 'Enrolled' status **and** there is an obligation y which is a 'TuitionFee' **and** that obligation is not paid (false), **then** x has a violation."*

FOL Statement

3d

$$\forall x\ [\ (\text{Student}(x) \land \text{StudentType}(x, \text{"Self-Support"}) \land \text{EnrollmentStatus}(x, \text{"Enrolled"}) \land \exists y\ (\text{HasObligation}(x, y) \land \text{ObligationType}(y, \text{"TuitionFee"}) \land \text{IsPaid}(y, \text{false}))) \rightarrow \text{Violation}(x, \text{"UnpaidTuition"})\ ]$$

FOL- PROVING THE USE CASES WITH FOL

Case A: Compliant Student (s1).

Goal: To prove that ¬Violation(s1)**.**

Facts about s1:

1. Student(s1)
2. StudentType(s1, "Self-Support")
3. EnrollmentStatus(s1, "Enrolled")
4. HasObligation(s1, ob1)
5. ObligationType(ob1, "TuitionFee")
6. IsPaid(ob1, true)

Rule 1: Tuition Fee Compliance (TuitionComplianceShape)

- ∀x [(Student(x) ∧ StudentType(x, "Self-Support") ∧ EnrollmentStatus(x, "Enrolled") ∧ ∃y (HasObligation(x, y) ∧ ObligationType(y, "TuitionFee") ∧ IsPaid(y, false))) → Violation(x, "UnpaidTuition")]

Rule 2: Accommodation Fee Overdue (AccommodationComplianceShape)

- ∀x, y, d [(Student(x) ∧ HasObligation(x, y) ∧ ObligationType(y, "AccommodationFee") ∧ OutstandingDuration(y, d) ∧ (d > 60)) → Violation(x, "OverdueAccommodation")]

Proof:

- **Check Rule 1:** The condition part of Rule 1 requires **IsPaid(y, false)**. For student **s1**, we know from **Fact 6** that **IsPaid(ob1, true).** Since one part of the AND condition is false, ***the entire condition is false.***

- **Check Rule 2:** The condition part of Rule 2 requires **ObligationType(y, "AccommodationFee"**). For student s1, we know from **Fact 5** that the obligation is a **TuitionFee**. ***This condition is false, so the rule does not apply.***

Conclusion: Since the conditions for a violation are not met for s1 under either rule, we can conclude that **¬Violation(s1) is False**

FOL- PROVING THE USE CASES WITH FOL

Case B: Unpaid Tuition Student (s2)

Goal: To prove that Violation(s2, "UnpaidTuition").

Facts about s1:

1. Student(s2)
2. StudentType(s2, "Self-Support")
3. EnrollmentStatus(s2, "Enrolled")
4. HasObligation(s2, ob2)
5. ObligationType(ob2, "TuitionFee")
6. IsPaid(ob2, false)

Rule 1: Tuition Fee Compliance (TuitionComplianceShape)

- ∀x [(Student(x) ∧ StudentType(x, "Self-Support") ∧ EnrollmentStatus(x, "Enrolled") ∧ ∃y (HasObligation(x, y) ∧ ObligationType(y, "TuitionFee") ∧ IsPaid(y, false))) → Violation(x, "UnpaidTuition")]

Rule 2: Accommodation Fee Overdue (AccommodationComplianceShape)

- ∀x, y, d [(Student(x) ∧ HasObligation(x, y) ∧ ObligationType(y, "AccommodationFee") ∧ OutstandingDuration(y, d) ∧ (d > 60)) → Violation(x, "OverdueAccommodation")]

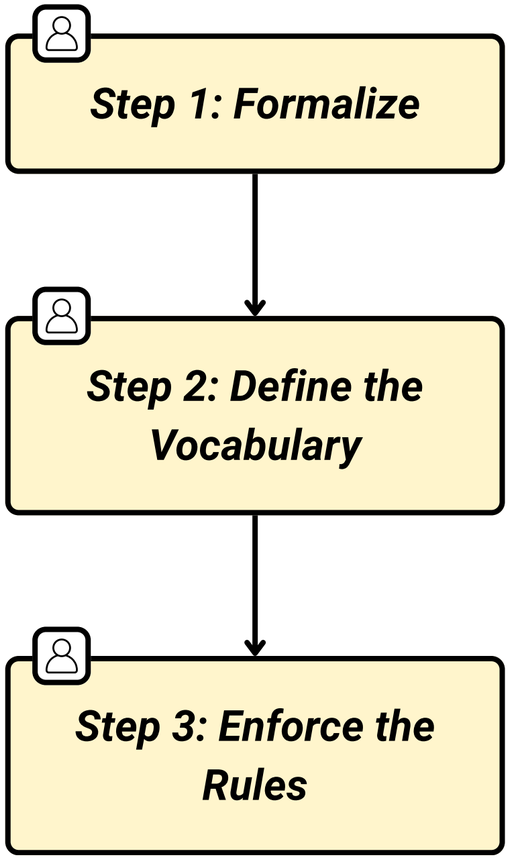
Proof:

- We consider **Rule 1: ∀x [(P(x) ∧ ∃y Q(x,y)) → Violation(x, r)]**
- We apply this rule to our specific student, s2.
- We check if the condition *(Student(s2) ∧ ... ∧ IsPaid(ob2, false)) is true.*
- We need to check the part ∃y (...). We can see that for y = ob2:
  - *HasObligation(s2, ob2) is true (Fact 4).*
  - *ObligationType(ob2, "TuitionFee") is true (Fact 5).*
  - *IsPaid(ob2, false) is true (Fact 6).*
- Since all parts of the condition are true, *the entire condition is true.*

Conclusion: We can logically conclude that ***Violation(s2, "UnpaidTuition") is true.***

POC - FOL TO ONTOLOGY & SHACL CONVERSION 4

- ***Practical Conversion Pattern***: *3 steps from FOL to Ontology and SHACL*



*Simplify natural language rules and convert them into a clear **FOL Statement.** This step will have extracted vocabulary, such as **Variables, Predicates and Proposition***

*Every **"Concept", "attribute"** and **"Relation"** defined in the Variables and Predicates section of FOL analysis must be formally created in the **Ontology.** (similar to data dictionary building)*

- ***Every Concept(variable) → Create a Class.***
- ***Every Relation(var1, var2) → Create a Property***
  - *Data Property*
  - *Object Property*

*The "**If...Then...**" (→) propositional structure of the FOL statement is converted into a **SHACL shape's "Target and Constraint"** structure.*

POC - FOL TO ONTOLOGY & SHACL

Step 1: Formalize

*Simplify natural language rules and convert them into a clear **FOL Statement.** This step will have extracted vocabulary, such as **Variables, Predicates** **and** **Proposition (Logical statement)***

Proposition
(logical statement)

- **P&P Statement:** "If the outstanding amount exceeds two month's rent, the student's status shall be 'suspended/dismissed'..."
- **FOL Statement:** *"For all students x, obligations y, and days d, **if** x has an obligation y **and** y is an 'AccommodationFee' **and** it is overdue by d days **and** d is greater than 60, **then** x has a violation."*

∀x, y, d [(Student(x) ∧ HasObligation(x, y) ∧ ObligationType(y, "AccommodationFee") ∧ OutstandingDuration(y, d) ∧ (d > 60)) → Violation(x, "OverdueAccommodation")]

Extracted Vocabulary

Variables

- **Student(x)**: x is a student.
- **Obligation(y):** y is a Obligation

Predicates
(Functional Predicate)

- **StudentType(x, t)**: x has the student type t (e.g., "Self-Support").
- **EnrollmentStatus(x, s)**: x has the enrollment status s (e.g., "Enrolled").
- **ObligationType(y, t)**: The obligation y is of type t (e.g., "TuitionFee", "AccommodationFee").
- **OutstandingDuration(y, d)**: The obligation y is overdue by d days.
- **Violation(x, r)**: x has a violation with the reason r.

Predicates
(Relational Predicate)

- **HasObligation(x, y)**: x has a financial obligation y.
- **IsPaid(y, b)**: The obligation y has a payment status of b (true/false).

POC - FOL TO ONTOLOGY & SHACL

Step 2: Define Vocabulary

*Every **"Concept", "attribute"** and **"Relation"** defined in the Variables and Predicates section of FOL analysis must be formally created in the **Ontology.** (similar to data dictionary building)*

- ***Every Concept(variable) → Create a Class.***
- ***Every Relation(var1, var2) → Create a Property***

Component from FOL Statement	Role in Ontology	Concrete Transformation	Result in Ontology (Turtle Syntax)
• ***Student(x)***	**Class**	The concept of a ***"Student"*** is used to create ***a primary class.***	ait:Student a rdfs:Class .
• ***ObligationType(y, t)***	**Class and Property**	The concept of an ***"Obligation" is*** created as ***another class*** *(ait:Obligation),* and its attribute ***"type" becomes a Data Property of that class.***	ait:Obligation a rdfs:Class . ait:hasObligationType rdfs:domain ait:Obligation .
• ***OutstandingDuration(y, d)***	**Data Property**	The ***"outstanding duration"*** attribute of an Obligation becomes a ***Data Property with a numeric value (integer).***	ait:outstandingDurationDays rdfs:domain ait:Obligation .
• ***HasObligation(x, y)***	**Object Property**	The ***"has obligation" relationship***, which connects a Student (x) to an Obligation (y), becomes an ***Object Property linking these two classes.***	ait:hasFinancialObligation rdfs:domain ait:Student ; rdfs:range ait:Obligation .

POC - FOL TO ONTOLOGY & SHACL

Step 3: Enforce Rules

*The "**If...Then...**" (→) propositional structure of the FOL statement is converted into a **SHACL shape's "Target and Constraint"** structure.*

Component from FOL Statement	Role in SHACL	Concrete Transformation	Result in SHACL Shape (Turtle Syntax)
• ***∀x (Student(x) ...)***	**Target**	***"For all students x"*** tells us that this SHACL shape will be applied to ***every instance of the ait:Student class.***	ait:AccommodationComplianceShape a sh:NodeShape ; sh:targetClass ait:Student ;
• ***(...) → Violation(...)*** *(The entire "If" part)*	**Constraint Condition**	Since we want to find cases that "violate" the rule, we will create a constraint that "searches for" the pattern that matches the entire "If" part of the FOL.	sh:sparql [...] ; *(In this case, a SPARQL-based constraint is most suitable due to the complex conditions.)*
• ***HasObligation(x, y) ∧ ObligationType(y, "AccommodationFee")***	**Graph Pattern**	The interconnected parts of the FOL become a graph pattern inside the ***WHERE*** clause of the SPARQL query.	$this ait:hasFinancialObligation ? obligation . ?obligation ait:hasObligationType "AccommodationFee" .

POC - FOL TO ONTOLOGY & SHACL

Step 3: Enforce Rules

*The "**If...Then...**" (→) propositional structure of the FOL statement is converted into a **SHACL shape's "Target and Constraint"** structure.*

Component from FOL Statement	Role in SHACL	Concrete Transformation	Result in SHACL Shape (Turtle Syntax)
• ***OutstandingDuration(y, d) ∧ (d > 60)***	**Filter Condition**	The value comparison part of the FOL becomes a ***FILTER*** clause in the SPARQL query.	?obligation ait:outstandingDurationDays ? days . FILTER(?days > 60)
• ***→ Violation(...)*** *(The "Then" part)*	**Constraint Outcome**	If the SPARQL query finds a pattern matching all the conditions ***(i.e., the SELECT $this WHERE { ... } returns a result)***, it means that the target node *($this)* does not conform to the shape and ***will be reported as a violation.***	sh:message "Student has accommodation fees overdue for more than 2 months (60 days)." ;

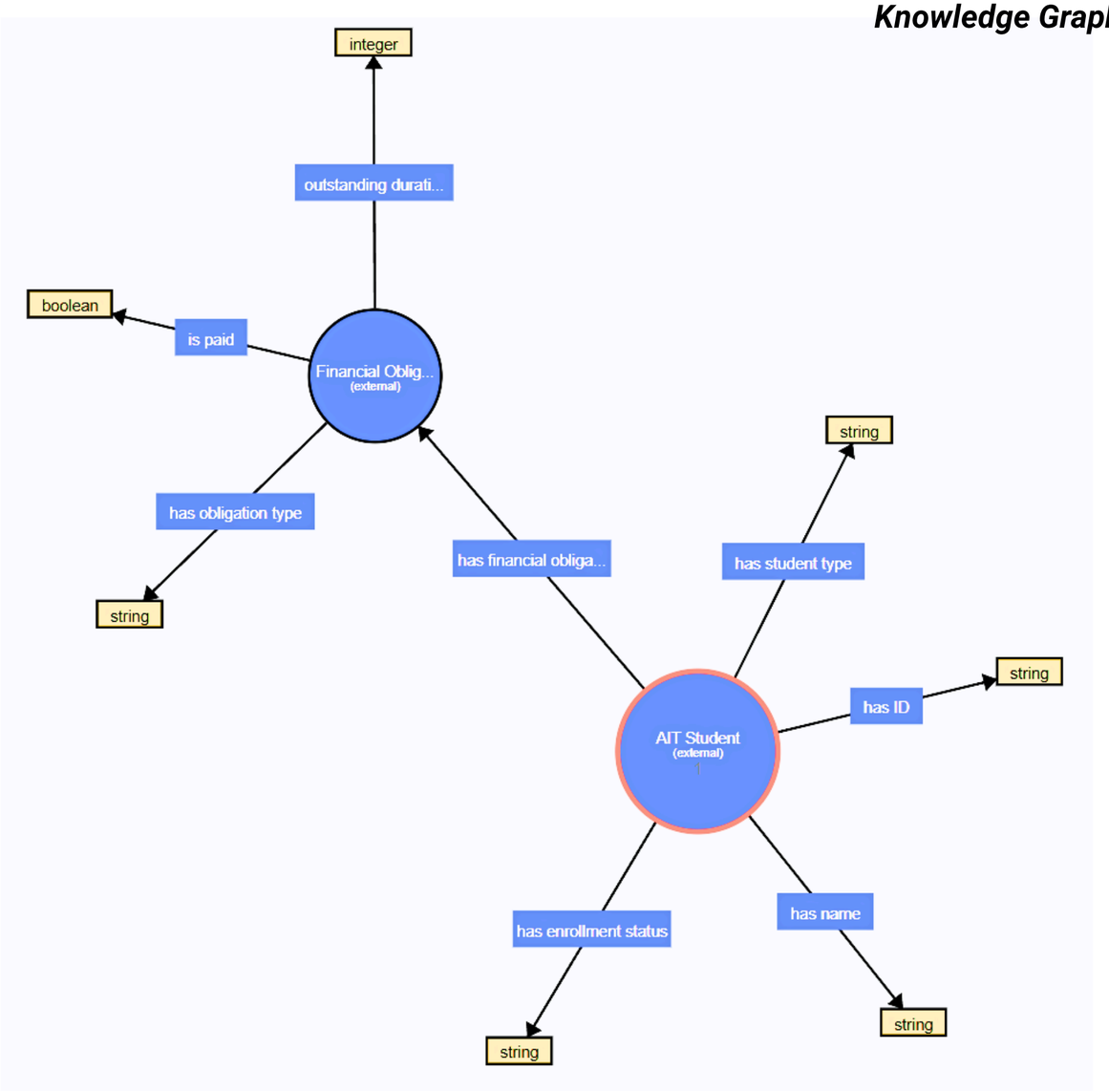
POC - SHAPE GRAPH (ONTOLOGY)

4

The result of FOL to Ontology and SHACL conversion

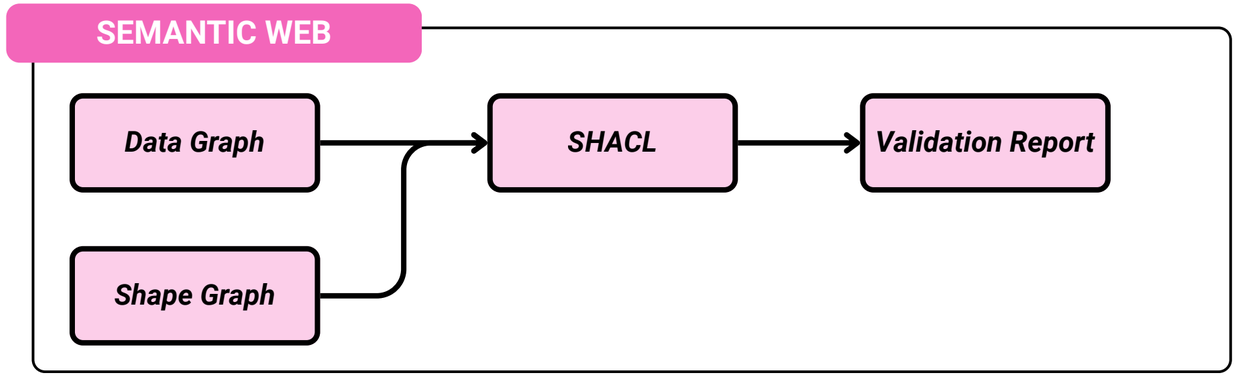
```
@prefix ait:   <http://ait.ac.th/ontology#> .
@prefix rdfs:  <http://www.w3.org/2000/01/rdf-schema#> .
@prefix rdf:   <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix xsd:   <http://www.w3.org/2001/XMLSchema#> .
@prefix owl:   <http://www.w3.org/2002/07/owl#> .

# Class Definitions
ait:Student
  a rdfs:Class ;
  rdfs:label "AIT Student"@en ;
  rdfs:comment "Represents a student enrolled at the Asian Institute of
Technology."@en .

ait:Obligation
  a rdfs:Class ;
  rdfs:label "Financial Obligation"@en ;
  rdfs:comment "Represents a financial obligation or invoice that a student
has."@en .

# Property Definitions
# -- Properties for Student Class
ait:hasId
  a rdf:Property ;
  rdfs:domain ait:Student ;
  rdfs:range xsd:string ;
  rdfs:label "has ID"@en ;
  rdfs:comment "The unique identifier for a student."@en .
.
.
```

Knowledge Graph



POC - SHACL VALIDATION

5



GraphDB

VALIDATION REPORT | DATA GRAPH

	Size	Uploaded	Imported	Context
caseA-compliantStudent .ttl Imported successfully in 1s. Added 4 statements	434 bytes	2025-09-26, 05:54	2025-09-26, 05:54	default
caseC-overdueAccommodation.ttl org.eclipse.rdf4j.sail.shacl.GraphDBShaclSailValidationException: Failed SHACL validation @prefix dash: <http://datashapes.org/dash#> . @prefix owl: ...	496 bytes	2025-09-25, 17:47	2025-09-25, 17:47	default
caseB-unpaidTuition.ttl org.eclipse.rdf4j.sail.shacl.GraphDBShaclSailValidationException: Failed SHACL validation @prefix dash: <http://datashapes.org/dash#> . @prefix owl: ...	451 bytes	2025-09-25, 17:47	2025-09-25, 17:47	default

NO Validation report

Validation report

Validation report

VALIDATION | SHAPE GRAPH

Imported successfully in less than a second.	2.16 kb	2025-09-25, 17:08	2025-09-25, 17:08	http://rdf4j.org/schema/rdf4j#SHACLShapeGraph
AIT-ontology.ttl Imported successfully in less than a second. Added 46 statements	2.82 kb	2025-09-25, 17:07	2025-09-25, 17:07	default

VALIDATION REPORT

CASE B

```
org.eclipse.rdf4j.sail.shacl.GraphDBShaclSailValidationException: Failed SHACL validation

@prefix dash: <http://datashapes.org/dash#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix rsx: <http://rdf4j.org/shacl-extensions#> .
@prefix rdf4j: <http://rdf4j.org/schema/rdf4j#> .
@prefix sh: <http://www.w3.org/ns/shacl#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .

_:42cbf76aafc748deb605e16215008a0048330 a sh:ValidationReport;
  sh:conforms false;
  rdf4j:truncated false;
  sh:result _:42cbf76aafc748deb605e16215008a0048331 .

_:42cbf76aafc748deb605e16215008a0048331 a sh:ValidationResult;
  sh:focusNode <http://ait.ac.th/ontology#student_652222>;
  rsx:shapesGraph rdf4j:SHACLShapeGraph;
  sh:value <http://ait.ac.th/ontology#student_652222>;
  sh:sourceConstraint _:fa0b6d7a-1488-4e5f-891e-0f1234ec2685-1;
  sh:sourceConstraintComponent sh:SPARQLConstraintComponent;
  sh:resultSeverity sh:Violation;
  sh:resultMessage "Self-Support students must pay tuition fees to be enrolled.";
  sh:sourceShape <http://ait.ac.th/ontology#TuitionComplianceShape> .

<http://ait.ac.th/ontology#TuitionComplianceShape> a sh:NodeShape;
  sh:targetClass <http://ait.ac.th/ontology#Student>;
  sh:sparql _:fa0b6d7a-1488-4e5f-891e-0f1234ec2685-1 .

_:fa0b6d7a-1488-4e5f-891e-0f1234ec2685-1 a sh:SPARQLConstraint;
```

CONFRIMATION

Validation Result: 1 item

Student Node

Message

Shape Name

CASE C

LLM required

```
org.eclipse.rdf4j.sail.shacl.GraphDBShaclSailValidationException: Failed SHACL validation

@prefix dash: <http://datashapes.org/dash#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix rsx: <http://rdf4j.org/shacl-extensions#> .
@prefix rdf4j: <http://rdf4j.org/schema/rdf4j#> .
@prefix sh: <http://www.w3.org/ns/shacl#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .

_:42cbf76aafc748deb605e16215008a0048517 a sh:ValidationReport;
  sh:conforms false;
  rdf4j:truncated false;
  sh:result _:42cbf76aafc748deb605e16215008a0048518 .

_:42cbf76aafc748deb605e16215008a0048518 a sh:ValidationResult;
  sh:focusNode <http://ait.ac.th/ontology#student_653333>;
  rsx:shapesGraph rdf4j:SHACLShapeGraph;
  sh:value <http://ait.ac.th/ontology#student_653333>;
  sh:sourceConstraint _:d32f1645-4895-46f9-897f-3e01736c3b78-2;
  sh:sourceConstraintComponent sh:SPARQLConstraintComponent;
  sh:resultSeverity sh:Violation;
  sh:resultMessage "Student has accommodation fees overdue for more than 2 months (60 days).";
  sh:sourceShape <http://ait.ac.th/ontology#AccommodationComplianceShape> .

<http://ait.ac.th/ontology#AccommodationComplianceShape> a sh:NodeShape;
  sh:targetClass <http://ait.ac.th/ontology#Student>;
  sh:sparql _:d32f1645-4895-46f9-897f-3e01736c3b78-2 .

_:d32f1645-4895-46f9-897f-3e01736c3b78-2 a sh:SPARQLConstraint;
```

CONFRIMATION

Validation Result: 1 item

Student Node

Message

Shape Name

EXPECTED RESULT

- *The proposed framework will achieve **high accuracy (defined as 100%)** for detecting violations of **low-complexity rules**. **For medium and high-complexity rules,** it is hypothesized that the framework will achieve **an accuracy of approximately 50%**, establishing a quantitative baseline for future research*

	Accuracy for Detecting Violations
Low-complexity rules	100%
Medium and High-complexity rules	50%

TENTATIVE WORK PLAN

	Jul-25				Aug-25				Sep-25				Oct-25				Nov-25				Dec-25				Jan-26				Feb-26				Mar-26				Apr-26			
Task	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Chapter1: Background & Introduction																																								
Chapter2: Literature Review																																								
Chapter3: Methodology																																								
- ***Phase1:*** *Proof of Concept*																																								
Proposal Presentation																																								
- ***Phase2:*** *Rule Selection*																																								
- ***Phase3:*** *FOL analysis*																																								
- ***Phase4:*** *Ontology & Development*																																								
- ***Phase5:*** *SHACL development*																																								
Chapter4: Experiment																																								
- *Demonstation & Evaluarion*																																								
Chapter5: Result																																								
Chapter6: Discussion & Conclusion																																								
Final Defense Presentation																																								